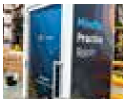

ZenBooths introduced by Amazon
ZenBooths will provide practices for health and mindfulness for warehouse employees.
http://dgit.in/jun21-09

Google launches Fuchsia
This is the third major operating system by Google after Android and Chrome OS.
http://dgit.in/jun21-10

Intel Core i9-11900K Desktop Processor

Not a worthy upgrade

Price ₹60,500

Intel's 11th Gen Core "Rocket Lake" family of desktop processors is the final stepping stone before Intel moves to their 10nm Superfin process in totality. We say 'in totality' because the 11th Gen Core desktop processors include both 14nm and 10nm based processors.

PERFORMANCE

In the single-threaded Cinebench benchmark, the Intel Core i9-11900K scores 626 points which is an improvement over the previous gen Intel Core i9-10900K. The AMD Ryzen 9 5950X is still ahead by a few points so AMD retains the single-threaded performance lead. The results of the multi-threaded run aren't surprising. Going from 10 cores in 10th Gen to 8 cores in 11th Gen does drop the performance of the 11900K below that of the 10900K. The AMD Ryzen 9 5950X, and 5900X are ahead but the Ryzen 8 5800X is very close to the 11900K.

POV-Ray being a multi-threaded benchmark, the 11900K falls behind the 10900K and all the higher core count AMD Ryzen 9 processors. However, in Blender the 11900K scores higher than the 10900K. The AMD Ryzen 9 5950X and 5900X are still higher and the Ryzen 7 5800X barely edges ahead. The Core i5-11600K falls behind the Ryzen 5 5600X by a narrow margin. Nevertheless, the algorithms used by Blender clearly favour the improvements brought in the 11th Gen processors.

We see that 7-Zip favours AMD's Zen 3 processors over the Intel 11th Gen processors. The scores seem similar to the other rendering benchmarks in our suite. So the 11900K falls behind the 10900K but the 11600K comes ahead of the 10600K. So core-for-core, the 11th Gen processors do show improvement but the lower core-count continues to be a handicap.

GAMING (DISCRETE GPU)

The scores here are for Shadow of the Tomb Raider, The Witcher 3, Assassin's Creed Odyssey and Strange Brigade. We've also averaged the FPS scores across the games. We can see that Intel is still the best for gaming with the 11900K leading the pack followed closely by the 11600K.

INTEL XE-LP IGPU PERFORMANCE

We see a 47 per cent improvement with the Intel Xe-LP over the older Intel Iris integrated graphics. If Intel had released Core i3 processors with Xe-LP graphics, then we'd have a great value offering in the entry-level segment. Compared to the AMD Ryzen 3 3200G, the Xe-LP with 32 EUs would be an improvement but compared to the AMD Ryzen 5 3400G, the Xe-LP would still lag behind. Running games on the iGPU also shows that we're seeing a similar 50 to 60 per cent improvement in average FPS. Unfortunately, current-gen AAA games aren't playable which is normal for all iGPUs but older AAA titles such as GTA V are playable on low settings.

80

PERFORMANCE..........84
VALUE FOR MONEY....75

INTEL ADAPTIVE BOOST TECHNOLOGY

We ran several benchmarks to see how Adaptive Boost Technology impacts performance. 3DMark Fire Strike scored 17442 without ABT and 17461 with ABT. The difference was within the margin or error. The Handbrake transcode workload took 12 minutes and 55 seconds with and without ABT. Video games also didn't show any improvements with ABT.

We logged the temperatures and noticed that the CPU Package temperature is always hovering near 70 degrees celsius, thus, Thermal Velocity Boost or Adaptive Boost Technology never actually kicked in during a sustained workload such as Handbrake. At this point, even with a really good CPU cooler in an air-conditioned environment, Thermal Velocity Boost and Adaptive Boost Technology aren't really helpful.

VERDICT

Intel seems to have made some significant improvements core-for-core in 11th Gen processors over 10th Gen processors. However, the Intel Core i9-11900K is not a worthy upgrade but the Intel Core i5-10600K is. And the integrated graphics is certainly a great improvement but it will have to get a little better to compete with AMD's Vega11 iGPU. All said, AMD's Ryzen 5000 series are still the better buy and we hope that changes with Intel's upcoming Alder Lake launch.

–Mithun Mohandas

SPECIFICATIONS
CORES: 8. | **THREADS:** 16 | | **BASE CLOCK:** 3.5 GHz | **BOOST CLOCK:** 5.3 GHz | **PROCESS NODE:** 14nm | **L3 CACHE:** 16 MB | **L2 CACHE:** 512 KB per core | **TDP:** 125 W.

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